

Computer Assisted Language Learning

ISSN: 0958-8221 (Print) 1744-3210 (Online) Journal homepage: www.tandfonline.com/journals/ncal20

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To cite this article: Ameneh Ghasemi & Mostafa Azari Noughabi (2026) Investigating the mediating role of Foreign Language Peace of mind and learning burnout in the relationship between Informal Digital Learning of English and willingness to communicate, Computer Assisted Language Learning, 39:3, 766-795, DOI: 10.1080/09588221.2024.2424321

To link to this article: https://doi.org/10.1080/09588221.2024.2424321

Published online: 18 Nov 2024.

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COMPUTER ASSISTED LANGUAGE LEARNING
2026, VOL. 39, NO. 3, 766–795
https://doi.org/10.1080/09588221.2024.2424321

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Investigating the mediating role of Foreign Language Peace of mind and learning burnout in the relationship between Informal Digital Learning of English and willingness to communicate

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ABSTRACT

Concurrent with the advancements of computer-assisted language learning (CALL), L2 researchers have started to pay more attention to the notion of informal digital learning of English (IDLE). A series of studies have already indicated its crucial role in linguistic achievements. However, the mediating role of positive and negative psychological states in the relationship between IDLE and willingness to communicate (WTC) has received scant research attention. Therefore, the present study explored whether foreign language peace of mind (FLPoM) and foreign language learning burnout (FLLB) mediated the relationship between IDLE and WTC among EFL learners. The participants were 316 Iranian EFL learners who completed four questionnaires. The results of structural equation modeling indicated that (a) IDLE did not have a direct impact on EFL learners' WTC, (b) FLPoM fully mediated the relationship between IDLE and WTC, and (c) the relationship between IDLE and FLLB was fully mediated by FLPoM. This study suggests that L2 learners' positive psychological state of FLPoM can direct their IDLE practices toward higher levels of willingness to interact in L2 and lower levels of negative syndrome of burnout. The findings of this study imply the significance of language learners' psychological states in their digital language learning in out-of-class learning environments.

ARTICLE HISTORY
Received 18 July 2023
Accepted 28 October 2024

KEYWORDS
Informal digital learning of English (IDLE); foreign language peace of mind (FLPoM); foreign language learning burnout (FLLB); willingness to communicate (WTC); EFL learners

Introduction

Learning English as a foreign language (EFL) has changed fundamentally in the wake of the Covid-19 pandemic. During the lockdown period, the so-called traditional face-to-face classes were replaced by online classes,

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and this gradually resulted in a shift from formal learning of English to informal learning of English with the use of digital devices (Nugroho & Atmojo, 2020). Informal digital learning of English (IDLE) refers to EFL learners' language practices outside the classrooms (e.g. chatting or commenting on social media in English, or watching English movies) with the help of different digital devices such as tablets, computers, or smartphones (Nugroho, 2021). Studies suggest that EFL learners have positive attitudes towards IDLE outside the classroom settings, and it can facilitate their language development (e.g. Nugroho & Atmojo, 2020). Furthermore, several studies investigated the relationship between IDLE and EFL learners' willingness to communicate (WTC) and found that the more EFL learners participate in IDLE practices, the higher their WTC in English (Lee et al., 2022; Lee & Dressman, 2018; Lee & Sylvén, 2021; Nugroho, 2021). Moreover, in addition to IDLE, which influences learners' WTC, an array of positive and negative psychological states can increase or decrease EFL learners' WTC (Lee & Chiu, 2023).

On one hand, there is a variety of learner emotional states (i.e. emotions) that have significant impacts on their L2 WTC (Lee & Hsieh, 2019). According to Alrabai (2022), positive and negative emotions predict L2 WTC; L2 learners who experience a positive emotion have greater WTC (MacIntyre & Vincze, 2017; MacIntyre & Wang, 2021), while those who experience a negative feeling have reduced levels of WTC (Khajavy et al., 2018). Other studies also show that emotional states mediate the effect of class social climate on learners' WTC (Wang et al., 2021). Accordingly, emotions are fundamental mediators of students' language learning achievements (Shao et al., 2019). Positive emotions (e.g. enjoyment, pride, and hope) can enhance students' motivation to learn a language while negative emotions (anger, boredom, and anxiety) are regarded to compromise language learning (Dewaele et al., 2022; Kruk et al., 2022; Pawlak et al., 2020, 2022; Shao et al., 2019). On the other hand, there is a wide range of psychological states (e.g. identity and motivation) that can determine learners' L2 WTC (Zarrinabadi et al., 2021). According to Cameron (2021), L2 learners' psychological states can diminish or enhance their L2 WTC; accordingly, negative psychological states (e.g. shyness) negatively affect how willing learners are to communicate in L2 (Chu, 2008), whereas positive psychological states (e.g. L2 self-confidence, and positive attitudes) positively correlate with learners' L2 WTC (Ghonsooly et al., 2012). Given the relevance of psychological states for students' WTC, exploring the mediating role of underexplored positive psychological states and negative psychological syndromes in language learning is essential.

Two underexplored psychological states in this regard are Foreign Language Peace of Mind (FLPoM) and Foreign Language Learning

Burnout (FLLB). FLPoM, a newly-introduced positive emotion, refers to how much someone feels comfort, coherence, and more importantly internal peace while L2 learning (Datu, 2017; Zhou et al., 2023). It can promote the conservation of students' personal resources due to its emphasis on emotional stability and harmony within and between individuals and their surroundings. This can consequently cause high stability of personal resources and low susceptibility to resource loss (Zhou et al., 2023). FLLB, a multi-dimensional syndrome, refers to emotional, physical, and mental exhaustion which is due to one's involvement in demanding and emotionally stressful conditions for an extended period (Li et al., 2021).

Despite the remarkable achievements in understanding the factors hindering or fostering L2 WTC, there are two underexplored issues in this intersection: The effects of online and computer-assisted learning environments on learners' L2 WTC (Zarrinabadi et al., 2021), and the mediating role of positive and negative psychological states in the relationship between IDLE and WTC. More specifically, to the best knowledge of the researchers, no previous research study has explored the mediating role of FLPoM and FLLB in the relationship between EFL learners' IDLE and their WTC. Therefore, the current study seeks to investigate this intersection. Informed by the tenets of positive psychology and as one of the first studies on the aforementioned variables in the field of foreign language education, this study can contribute to the literature by proposing a mediation model of the role of IDLE in foreign language learners' WTC.

Literature review

Informal Digital Learning of English

Growing up with a wide range of different technologies, today's generation socialize in a digital world, and they can generate English information anywhere and anytime, thereby having numerous chances to practice English. According to Richards (2015), this can transform the way of language learning. With this drastic change in learning conditions, more attention has been paid to informal language learning (e.g. Benson, 2011; Sun et al., 2017). Nowadays, an increasing number of learners tend to learn and practice English in informal, digital environments (Lee, 2019; Zhang & Liu, 2022). Benson (2011) introduces a four-dimensional explanatory framework corresponding to the basic characteristics of out-of-class computer assisted language learning (CALL) which is also applicable to IDLE. The four dimensions of this framework include (a) formality (informal, formal, or non-formal learning), (b) location (extramural, extracurricular, out-of-class, and in-class), (c) pedagogy (self-instructed, instructed, or naturalistic), and (d) locus of control

(other-directed or self-directed). According to Lee (2019), IDLE is a self-directed and autonomous English learning by using a wide range of digital technologies including internet websites, social media, and smartphones. Research shows that IDLE may be affected by a passion for language learning because enthusiastic learners are more likely to love informal English learning, which can improve their speaking ability and communicative skills (Dressman, 2019; Lee et al., 2022). Furthermore, Lee and Lee (2021) define IDLE as using English in informal digital settings without being controlled or evaluated by a teacher due to one's interest. Thus, it is evident that digital technologies can offer opportunities for EFL students to learn and practice a foreign language (Azari Noughabi & Ghasemi, 2024; Lee, 2019; Soyoof et al., 2023). In the current study, IDLE refers to EFL learners' use of different digital devices to learn and practice the English language in an informal setting outside the classrooms without being assessed by a teacher.

There are two main categories of IDLE activities, namely productive and receptive IDLE activities. Receptive IDLE activities refer to EFL learners' attempts to improve their L2 input, L2 vocabulary, and topic knowledge (e.g. watching English movies, listening to English podcasts, or reading English newspapers). As Ellis and Wulff (2014) state, language learning primarily depends on learners' engagement with the linguistic input they receive. According to Long's (1996) revised interaction hypothesis, L2 input can be more comprehensible to learners when they selectively pay attention to L2 structures and vocabulary and negotiate meaning. Therefore, the role of the linguistic environment in increasing learners' attention to L2 input is undeniable.

Productive IDLE activities refer to EFL learners' attempts to produce L2 output and interact with English speakers (e.g. sending emails in English, posting comments in English, or chatting in English). Output is produced when a series of processes such as access and production procedures are involved to interact through proper forms and structures (Gass et al., 2020). According to Gass and Mackey (2014), the process of second language acquisition is dynamic and the relationship between input, interaction, and output is circular and interdependent. Comprehensible input can be transformed into intake through interaction, defined as conversations that students have (Gass & Mackey, 2020). L2 output, which can be promoted through interaction, can also act as a foundation of new L2 input (Mitchell et al., 2013). As stated by Swain (1995), L2 output serves the function of enhancing the comprehensibility of L2 input in the process of language learning through interaction and timely feedback. Therefore, learners' communicative competence is affected by receiving comprehensible L2 input and producing valid L2 output (Swain, 1985; Van Patten et al., 2020). Considering the

interactionist theory of SLA, Fuente (2003) argued that computer-based, interactive, communicative tasks promoted linguistic outcomes and language proficiency. Accordingly, receptive and productive IDLE activities can enhance language learners' motivation to communicate in L2 (Lee & Drajati, 2019).

One line of inquiry has been devoted to the investigation of the role of IDLE in linguistic achievements. In a recent study, Lee (2019) investigated how IDLE activities were associated with L2 vocabulary learning outcomes among 77 Korean EFL university students (22 males, and 55 females, ages 19–27) who majored in English and non-English majors. The study of Lee (2019) revealed that a balance between the quantity and diversity of meaning- and form-focused IDLE practices and teachers' affective support in engaging with technology-enhanced activities could facilitate L2 vocabulary learning. Odinokaya et al. (2021) investigated the effectiveness of using the Discord application (a free messenger with support for Internet Protocol telephony, conferencing, public and private chats, and text and voice message exchange) in facilitating EFL learners' vocabulary acquisition. Their study was conducted in a university in Russia and lasted three months. The participants were 80 Russian EFL students (ages 17–19) majored in Intercultural Communication and Foreign Languages/Linguistics majors. All of the participants were at an intermediate level of English proficiency. The investigation found that utilizing Discord as a complementary tool for teaching vocabulary could enhance EFL learners' vocabulary gains and further facilitate using new words in their speaking.

A series of studies explored the role of online learning platforms in language learning. Yen et al. (2015) conducted a case study on 42 EFL students (10 males and 32 females) with an average age range of 16, enrolled in an English conversation course in a business college in Taiwan. The findings revealed that using Facebook and Skype for asynchronous and synchronous online discussions improved EFL learners' speaking and writing skills through role-playing learning activities. Wongsa and Son (2022) explored the effect of drama-based learning activities and Facebook on Thai secondary school EFL learners' speaking skills and motivation in language learning. The participants were 40 Thai EFL students from two Grade 12 classes as well as two Thai EFL teachers at a high school in Thailand. They found that drama-based activities and online language learning through Facebook positively influenced EFL learners' speaking skills and motivation.

In a mixed-methods study, Lee and Drajati (2019) examined the relationship between affective variables, willingness to communicate, and IDLE activities among 324 Indonesian undergraduate and postgraduate EFL university students (70 females and 254 males) majoring in the English

language (with an average age of 21 years old). The results indicated that productive IDLE practices were significant predictors of cross-cultural communication strategies, while receptive IDLE practices were significant predictors of positive perception of English varieties. Moreover, Lee and Drajati (2019) found that only productive IDLE activities, grit, self-confidence, and motivation significantly predicted their WTC. The study by Lee and Drajati (2019) suggests that learners' engagement with IDLE activities plays a key role in facilitating L2 communication. Dressman (2019) conducted a three-year study to explore how Moroccan students learn English through digital resources in public universities. The participants were 107 male and female students from three universities in Morocco, and the majority of them were first-semester students. Dressman (2019) found that students typically had access to digital resources (e.g. television programs and live interactions with tourists) outside classroom instruction which helped them achieve higher speaking scores.

In a recent study, Nugroho (2021) investigated 156 Indonesian EFL learners' IDLE practices and portrayed the existence of a significant relationship between EFL learners' IDLE and their WTC in English. The participants (63 males and 93 females, age range: 17–20) had enrolled in English language studies at a public university in Indonesia. The investigation revealed that the participants were willing to communicate when participating in various IDLE (productive and receptive) activities. This means the extent to which EFL learners were engaged in IDLE practices was a predictor of their WTC in L2. Lee et al. (2022) investigated how foreign language enjoyment (composed of personal enjoyment, social enjoyment, and teacher appreciation) mediated the relationship between IDLE and WTC among 308 EFL Hong Kong secondary students (ages 12–18) inside and outside the classroom. The results showed that all the subcomponents of FLE mediated the relationship between IDLE and L2 WTC inside the classroom setting.

Their study also found that personal and social enjoyment significantly mediated the link between IDLE and L2 WTC outside the classroom. Lee et al. (2022) concluded that the more intensively EFL learners were engaged in productive and receptive IDLE activities, the more willing they were to communicate in English. This shows that engagement in IDLE practices can increase EFL learners' motivation to be involved in communication in a foreign language. However, in a recent study on 246 Kazakhstani EFL university students in Kazakhstan, Zadorozhnyy and Lee (2023) found that IDLE was not directly associated with WTC, and they identified self-efficacy as a mediator of this relationship. In sum, the available findings are inconclusive and further evidence is needed to understand the relationship between IDLE and learners' willingness to communicate in L2.

Willingness to communicate

Over the past years, learners' WTC has been believed to be an influential individual difference (ID) factor in facilitating L2 acquisition (MacIntyre et al., 1998; Peng & Woodrow, 2010). Initially, McCroskey and Baer (1985) proposed the construct of WTC in the mother tongue (L1). They defined it as a 'personality-based, trait-like predisposition which is relatively consistent across a variety of communication contexts and types of receivers' (p. 6). After a decade, MacIntyre et al. (1998) distinguished L1 WTC from L2 WTC and defined L2 WTC as 'a readiness to enter into discourse at a particular time with a specific person or persons, using a L2' (p. 547). According to the heuristic model of L2 WTC, proposed by MacIntyre et al. (1998), a multitude of individual, contextual, and sociocultural variables can influence language learner' L2 WTC. Accordingly, L2 WTC is a multi-faceted variable affected by a wide range of variables including motivation, self-confidence, and achievement goals (Feng et al., 2023; Peng & Woodrow, 2010; Yashima, 2019). In the present study, WTC refers to the extent to which EFL learners are willing to interact and communicate in English at a particular time.

In line with the proliferation of positive psychology studies in SLA (MacIntyre et al., 2019), identifying the links between learners' emotional and psychological states and WTC has been the focus of recent studies. Kun et al. (2020) investigated the relationship between L2 WTC and foreign language enjoyment (FLE) among 128 EFL university students (male = 51, and female = 77) in China. The participants were first- and second-year students in various non-English majors. The investigation suggests a moderately strong positive relationship between L2 WTC and FLE. Furthermore, Khajavy et al. (2021) explored the relationship between WTC, foreign language anxiety (FLA), and FLE among 38 Iranian EFL university students (11 males and 27 females, ages 18–22) who majored in Teaching English as a Foreign Language (TEFL). It was found that WTC and FLE were consistently and positively correlated. However, WTC and anxiety, as well as anxiety and FLE were inconsistently and negatively correlated. In a recent study, Feng et al. (2023) explored the link between 408 Chinese secondary school students' WTC and their achievement goals (mastery goals, performance-approach, and performance-avoidance goals) *via* enjoyment and anxiety. Their study revealed that mastery and performance-approach goals were positively related to learners' WTC *via* enjoyment. At the same time, anxiety did not play a significant role in the link between achievement goals and L2 WTC.

Lee et al. (2024) investigated the mediating roles of FLE and FLA in the relationship between IDLE and L2 WTC among 1,265 Korean EFL

learners (856 females, and 400 males; 501 tertiary and 764 secondary level students; ages 14–27) from one university and ten secondary schools in two Korean cities. The results showed that both variables were partial mediators in the relationship between IDLE and WTC, with FLE being a stronger mediator than FLA. Another empirical investigation on the mediating role of boredom in the relationship between IDLE and WTC is the longitudinal study of Taherian et al. (2023) involving 354 Iranian EFL university students (136 males and 218 females, ages 18–33) attending online general English courses at three Iranian universities. All the participants were English-major students who had lower-intermediate to upper-intermediate English proficiency. The investigation showed that foreign language boredom significantly mediated the developmental relationship between Iranian EFL learners' IDLE and WTC in L2.

Highlighting the potential benefits of gameplay for language learning purposes, Reinders and Wattana (2015) explored the impact of digital game-based language learning on 30 Thai EFL students' linguistic practices and WTC in English. The participants had different English proficiency levels, and all of them were third-year university students majoring in IT. They were enrolled in a 15-week English for Information Technology course with six units focusing on developing learners' linguistic skills and practical English communication. At the end of each unit, the students played Ragnarok Online (a massively multiplayer online role-playing game). They completed various quests (i.e. an assigned mission in a game that players try to accomplish). The game had been specifically modified based on the students' learning context and course materials. Reinders and Wattana (2015) found that the gameplay could lower language learners' affective obstacles to language learning and enhance their L2 WTC by encouraging L2 interaction. It is also believed that informal English learning through gamified mobile applications (e.g. DuoLingo) affects learners' motivation (Shortt et al., 2023). It has been shown that IDLE practices enhance learners' motivation (Lee & Lee, 2021; Liu et al., 2023; Soyoof et al., 2023) which has been recognized as a major determinant of WTC in L2 (Alrabai, 2022). Therefore, motivated functioning in online language learning communities and using gameplay for language learning purposes can mediate the relationship between WTC and digital English learning by lowering learner' affective barriers (Reinders & Wattana, 2015).

Foreign Language Learning Burnout

The complex construct of foreign language learning burnout (FLLB) is a multi-faceted psychological syndrome in L2 learning, which is typically

characterized by mental exhaustion, restricted efficacy, and cynicism. However, little attention has been paid to learner burnout in EFL contexts (Li, 2020). Vizoso et al. (2019) define burnout as a 'psychological syndrome characterized by gradual emotional fatigue, loss of motivation, and reduced enthusiasm' (p. 768). Li et al. (2021) state that taking actions to counter FLLB contributes to L2 learners' language development, mental health, and well-being. The importance of exploring FLLB is also in line with the perspectives of positive psychology that aims to increase language learning outcomes and improve L2 learners' well-being (Li, 2020). In this study, similar to the study of Li et al. (2021), FLLB corresponds to Maslach et al.'s (2001) model of burnout that encompasses three kernel dimensions, including emotional exhaustion, cynicism (or depersonalization), and inefficacy. The first dimension, i.e. emotional exhaustion, is a state of severe exhaustion and fatigue due to the draining of emotional resources. The next dimension, i.e. cynicism, refers to a state of having indifferent, distancing, or negative attitudes towards one's work and whoever involved. The last dimension, i.e. inefficacy, is characterized by feeling underachieving or incompetent at work. Maslach et al.'s (2001) model shows that burnout is more than an emotion. Burnout is regarded as a multi-faceted concept that includes three main dimensions: exhaustion, cynicism, and inefficacy.

Li (2020) conducted a correlational study on 1,307 Year-2 senior high school EFL students' burnout in China and reported burnout as an influential and negative predictor of L2 learners' exam performance and self-rated language proficiency. Furthermore, Liu et al. (2021) investigated the relationship between FLLB and self-oriented perfectionism under the influence of language learning anxiety and grit among 544 Chinese EFL college students (468 females and 76 males, ages 17–28). The study showed the indirect impact of perfectionism on students' English learning burnout in mobile learning contexts through their grit and learning anxiety. After all, examining the role of L2 learners' various psychological states in their L2 WTC using a holistic perspective of positive psychology is needed, and this perspective is gaining more attention in SLA studies (Dewaele & MacIntyre, 2014; Li, 2020). Nevertheless, up to now, few research attempts have targeted the multi-dimensional concept of FLLB and the novel concept of foreign language peace of mind and their mediating role in linguistic outcomes (Li et al., 2021; Zhou et al., 2023).

Foreign Language Peace of Mind

The emergence of positive psychology in SLA has triggered interest in studying the role of positive emotions in language learning (MacIntyre

et al., 2019; Yang et al., 2023). Khajavy et al. (2018) state that WTC models lack direct incorporation of positive emotions. The concept of peace of mind (PoM) was initially introduced by Lee et al. (2013) who conceptualized this new construct as a state of inner peace and harmony. Inner peace is associated with a low-arousal positive (LAP) emotional state, and harmony is linked to a state of balance within a person and between that person and his or her surroundings. The two dimensions of PoM (i.e. inner peace and harmony) are integrated into a coherent, homogeneous process (Lee et al., 2013). PoM entails a well-regulated, homeostatic, and stable state of mind (Zhou et al., 2023). Zhou et al. (2023) introduced the concept of foreign language peace of mind (FLPoM) as a new positive emotion in the Chinese EFL context based on Chinese cultural orientations. FLPoM can expand a learner's momentary though-action repertoire by focusing on the activity and minimizing external distractions (Li et al., 2021); this is in line with Fredrickson's (2001) broaden-and-build theory, a theoretical perspective on positive emotions that situates within the field of positive psychology. Fredrickson's (2001) broaden-and-build theory argues that experiencing positive emotions can 'broaden people's momentary thought–action repertoires, which in turn serves to build their enduring personal resources, ranging from physical and intellectual resources to social and psychological resources' (p. 218). Informed by broaden-and-build theory and positive psychology, Badiei et al. (2023) found that positive emotions (e.g. FLE and FLPoM) can enhance EFL learners' proficiency level. Underpinned by broaden-and-build theory, the study of Ebn-Abbasi et al. (2024) showed that positive variables (i.e. L2 grit, growth mindset, and ideal L2 self) could increase EFL learners' WTC.

The majority of studies on PoM have been conducted in the field of educational psychology. Datu (2017) reported positive relationships between 525 Philippian high school students' PoM, their academic achievement, controlled motivation, and autonomous motivation. The participants were 257 boys and 268 girls (ages 11–19). Furthermore, Datu et al. (2018) conducted a study on 606 Filipino high school students (300 males, 305 females, and one participant who failed to mention his/her gender) with an average age range of 13.87. The investigation found a positive relationship between academic engagement and PoM. In a pioneering study on foreign language learners' PoM, Zhou et al. (2023) designed a domain-specific scale to measure EFL learners' PoM. The participants were two independent samples that comprised 599 high school students in China. The first sample included 158 EFL high school students (112 females and 46 males, ages 15–21), and the second sample included 441 EFL high school students (280 females and 161 males; ages 15–21). It was found that FLPoM was a stronger predictor of the

participants' self-perceived language proficiency level compared to FLE. In a recent pseudo-longitudinal study, Dewaele and Meftah (2023), investigated the trajectory of two positive emotions (FLE and FLPoM) and two negative emotions (foreign language classroom anxiety and foreign language boredom) among 502 Moroccan EFL learners (269 female and 228 males; mean of age = 22.4 years old) with beginner, intermediate, and advanced English proficiency levels from different language centers, universities, and secondary schools in Morocco. Their study revealed that FLE and FLPoM increased significantly across skill levels. Dewaele and Meftah (2023) also found that increased skill mastery yielded a greater maintenance of inner harmony among EFL learners.

Overall, reviewing the literature reveals lack of conclusive results regarding the influence of IDLE on EFL learners' positive and negative psychological states. This issue might be attributed to the fact that research on IDLE is still young (Azari Noughabi & Ghasemi, 2024; Lee, 2019). Earlier research on FLPoM has reported its positive role in facilitating foreign language learning (Badiei et al., 2023; Zhou et al., 2023). However, due to the novelty of the concept of FLPoM in the field (Chen et al., 2023; Zhou et al., 2023), scant research attention has been paid to exploring its mediating role in the relationship between EFL learners' IDLE and WTC. While previous research reveals that learning burnout is influenced by different factors such as anxiety (Liu et al., 2021) and negatively affects foreign language learning (Li et al., 2021), exploring whether FLLB mediates the relationship between digital English learning and L2 WTC has remained under-explored. According to Teimouri (2016) and Dewaele et al. (2023), exploring the role of both positive and negative psychological states in the language learning process and linguistic outcomes is significant.

Previous research indicates that learners' digital competence is positively associated with the deep learning approach and negatively associated with burnout in digital EFL learning (Niu et al., 2022). Additionally, according to Kumpikaitė-Valiūnienė et al. (2021), social aspects of digital competencies can help students deal with stress and burnout in online learning. Moreover, cynicism has negatively affected EFL students' learning outcomes (Niu et al., 2022). Learning burnout is positively related to weak digital competence, high technostress, and low learning outcomes among EFL learners (Niu et al., 2022). Therefore, it might be hypothesized that burnout can mediate the relationship between digital EFL learning and students' L2 achievements and WTC (Lee et al., 2024; Lee & Sylvén, 2021; Niu et al., 2022). Therefore, the current study aims to add evidence to this new research line by investigating whether FLPoM and FLLB mediate the relationship between IDLE and WTC.

In sum, considering the literature review, we hypothesize that IDLE is linked to learners' WTC since it has been reported in a few EFL contexts that L2 learners' engagement in IDLE practices enhances their L2 WTC (e.g. Lee & Drajati, 2019; Lee & Dressman, 2018; Lee & Sylvén, 2021; Nugroho, 2021). We also hypothesize that IDLE is linked to FLPoM and FLLB (Lee et al., 2022). According to Lee and Lee (2021), IDLE significantly impacts learners' positive emotions (i.e. FLE). Learners engaged with English learning in out-of-class digital settings are more likely to promote their ideal L2 self-images, experience fewer negative psychological states (e.g. exhaustion and burnout), and enjoy learning a foreign language. Given that FLE and FLPoM are highly correlated (Zhou et al., 2023), it might be hypothesized that students' engagement in IDLE activities may influence their FLPoM as well. FLPoM and FLLB are hypothesized to be linked to WTC, since according to previous research, positive and negative learner variables predict learners' L2 WTC (e.g. Alrabai, 2022; Ebn-Abbasi et al., 2024; Khajavy et al., 2018; Lee & Hsieh, 2019; Wang et al., 2021). Finally, we hypothesize that FLPoM and FLLB mediate the relationship between IDLE and WTC; this hypothesis is driven based upon the previous research that showed the mediating role of several positive and negative learner variables including foreign language enjoyment, foreign language anxiety, and foreign language boredom in the relationship between IDLE and L2 WTC (e.g. Lee et al., 2024; Taherian et al., 2023). The hypothesized paths are shown in Figure 1.

Method

Participants

A total of 316 EFL learners from different provinces of Iran constituted the sample of this study. Their age ranged from 15 to 52 ($M = 27.38$, $SD = 6.20$). They were learning English as their foreign language in different contexts, such as private language institutes, public schools, or universities. The majority of the participants had been learning English in the university context (76%) and had educated in English Language

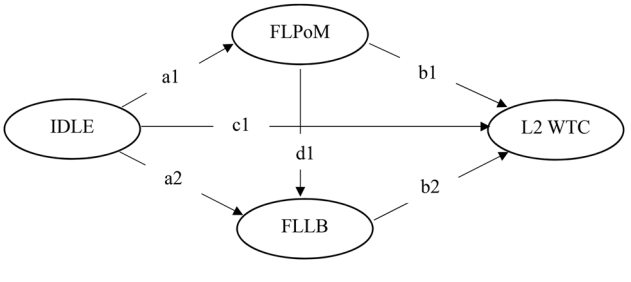


Figure 1. The theoretical mediation model.

Teaching (66%). The participants were selected through snowball sampling. Table 1 presents a more detailed overview of the demographic characteristics of the sample. Informed consent was obtained from all the participants.

Instruments

Informal digital learning of English scale

To measure EFL learners' IDLE, the Informal Digital Learning of English Scale, developed by Lee and Drajati (2019), was used. The scale has 13 items on two factors: receptive IDLE activities (e.g. 'I play games in English.') and productive IDLE activities (e.g. 'I send an email to others in English.'). Items are rated on a 5-point Likert scale, ranging from 1 (*never*) to 5 (*very often/many times per day*). The study of Lee and Lee (2021) confirmed the reliability and convergent validity of the Informal Digital Learning of English Scale (α = .82; Composite Reliability = .80). In this study, the Informal Digital Learning of English Scale had an internal consistency of α = .87.

L2 Willingness to communicate scale

In this study, the L2 Willingness to Communicate Scale, developed by Peng and Woodrow (2010), was used to measure learners' L2 WTC. The scale has a one-factor structure with ten items (e.g. 'I am willing to give a short self-introduction without notes in English to other people.'), and the items are rated on a 5-point Likert scale ranging from 1 (*definitely not willing*) to 5 (*definitely willing*). Peng and Woodrow (2010) revealed that their scale was both valid and reliable (α = .89, GFI = .97, CFI = .98, RMSEA = .07). In the current study, the reliability index of the L2 Willingness to Communicate Scale estimated *via* Cronbach's alpha was .86.

Table 1. Demographic information about the participants.

Variable	Frequency
Gender	Female = 234
	Male = 82
Learning Context	Private Language Institutes = 57
	Public Schools = 17
	Universities = 242
Academic Degree	Diploma = 22
	Bachelor of Arts = 119
	Master of Arts = 151
	Ph.D. = 22
	Other = 2
Major	English Language Teaching = 211
	English Language Literature = 34
	English Translation Studies = 52
	Other = 19

Foreign language learning burnout scale

The Foreign Language Learning Burnout Scale, designed by Li et al. (2021), was used to measure the participants' FLLB. The scale has ten items on a three-factor structure including exhaustion (e.g. 'I feel emotionally drained by my English studies.'), cynicism (e.g. 'I have become less enthusiastic about my English studies.'), and reduced efficacy (e.g. 'I don't feel stimulated when I reach my goals in English study.'). Items are rated on a 5-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). As reported by Li et al. (2021), this scale is a reliable and valid measure of FLLB (exhaustion α = .88, cynicism α = .83, reduced efficacy α = .81, CFI = .97, TLI = .96, RMSEA = .06). In the present study, a Cronbach's alpha reliability coefficient of .89 was reported for the Foreign Language Learning Burnout Scale.

Foreign language peace of mind scale

To assess EFL learners' peace of mind, the Foreign Language Peace of Mind Scale, developed by Zhou et al. (2023), was used. The scale has a one-factor structure with eight items (e.g. 'My mind is free and at ease in the English class.'), and it is rated on a 5-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). The study of Zhou et al. (2023) confirmed that this scale has high internal consistency and a valid factor structure (Cronbach alpha = .95, CFI = .98, TLI = .98, RMSEA = .07). In the present study, the reliability of the Foreign Language Peace of Mind Scale estimated *via* Cronbach's alpha was .91.

Data collection

The data for this study were collected from May 2023 to June 2023 from Iranian EFL learners. No deadline was set for the participants to complete the survey. To include samples from different areas of the country, the Google Forms platform was utilized to collect data. Snowball sampling, which is a form of non-probability sampling, was used to recruit the participants. The link of the electronic survey was shared with the participants of this study through social media applications, and the participants were kindly invited to participate in the research. The survey also included a separate section providing instructions on how to complete the forms. Before doing so, the participants were initially expected to sign the informed consent form while they were assured that all data would be kept anonymous and confidential.

Data analysis

After data collection, the data were initially analyzed to ensure data normality and a lack of missing data. The initial analysis, which involved Pearson's correlation coefficients and scale internal consistencies, was calculated in SPSS version 26. To create a single observed score for each latent variable, the average scores were calculated for each measurement instrument. The subscales of each latent variable were taken into account. Given that mediation analysis is regarded as a family part of the structural equation modeling (Kline, 2015), the mediation model was analyzed in AMOS 24. The goodness-of-fit indices were examined to check whether the model fits the data. The chi-square/degree of freedom ratio (χ^2/df), the comparative fit index (CFI), the goodness of fit (GFI), the root-mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR) were taken into account to check the model fit (Hu & Bentler, 1995). A mediation model was fitted with IDLE as the predictor variable, FLPoM and FLLB as the mediators, and L2 WTC as the outcome variable. To estimate the mediation model, a maximum likelihood estimator and a 5000-sample bootstrapping with a 95% confidence interval were employed (Biesanz et al., 2010). Figure 1 depicts the theoretical mediation model of the present study.

The direct (c1) and indirect (a1 + b1; a2 + b2; a1 + d1) effects were examined. The mediation was indicated in the case where indirect effect estimates were significant. In addition, if the direct effects (c1) were reduced to zero after adding the mediators to the model, the relationship between the predictor (IDLE) and the outcome variable (L2 WTC) was determined to be fully mediated by the mediators (FLPoM and FLLB). On the other hand, when both the direct and indirect effects were statistically significant, a partial mediation was achieved.

Results

Descriptive statistics and correlation coefficients

The descriptive statistics and the skewness and kurtosis of each variable of each variable are shown in Table 2. As indicated in Table 2, the data were normally distributed.

Next, the Pearson correlation coefficients between the variables were calculated to measure the strength of the linear relationships between the main variables. As shown in Table 3, all correlations were significant. In addition, the correlation matrix indicated that no correlation was too large to cause multicollinearity ($r > .80$; Field, 2013). The magnitude of effect sizes was higher than the average (.69) based on Plonsky and Oswald's (2014) suggested criteria for L2 research.

Table 2. Descriptive statistics and normality distribution.

	Minimum	Maximum	Mean	Std. Deviation	Skewness	Kurtosis
FLPoM	1.25	5.00	3.716	.802	−0.372	−0.392
L2 WTC	1.40	5.00	3.686	.762	−0.269	−0.569
FLLB	1.00	4.10	2.015	.778	.647	−0.384
IDLE	1.15	4.77	2.702	.777	.312	−0.325

Table 3. Correlation Matrix.

Variable	1	2	3	4
FLPoM	–	.441**	−0.452**	.164**
L2 WTC		–	−0.300**	.212**
FLLB			–	−0.139*
IDLE				–

Note. $^{**}p < .01$, $^{*}p < .05$.

Table 4. The goodness-of-fit indices for the variables.

	χ^2/df	GFI	CFI	RMSEA
FLPoM	2.38	.96	.98	.06
FLLB	2.33	.95	.97	.06
L2 WTC	3.09	.90	.91	.08
IDLE	3.01	.90	.90	.08
Acceptable range	<3	>.90	>.90	<.08

Confirmatory factor analysis

CFA was conducted to ensure the construct validity of the scales. The factor loadings for each variable were all high enough (>.50) except for Item 3 of the L2 WTC. Accordingly, we excluded this item from the measurement model. The estimated AVEs were beyond 0.4, ensuring a good discriminant validity. Moreover, the measures of each variable showed a high internal consistency (estimated through Cronbach's alpha). The goodness-of-fit indices for each main variable are shown in Table 4.

Mediation model

To recognize the mediation role of FLLB and FLPoM in the relationship between IDLE and L2 WTC, the theoretical mediation model was tested. The full mediation model is visually shown in Figure 2. The results indicated a good model fit; χ^2/df = 2.00, GFI = .90, CFI = .94, RMSEA = .05, SRMR = .08.

The results of mediation model analysis, shown in Table 5, revealed that two of the indirect effects of the mediation model were statistically significant. Specifically, FLPoM mediated the relationship between EFL learners' IDLE and WTC (β = .032; $p < .01$, CI [.006; .085]) and FLLB (β = −0.077; $p < .01$, CI [−0.165; −0.011]). Accordingly, when FLPoM entered the mediation model, a full mediation was indicated. In addition, it was found that IDLE did not influence L2 WTC directly (β = .165; p

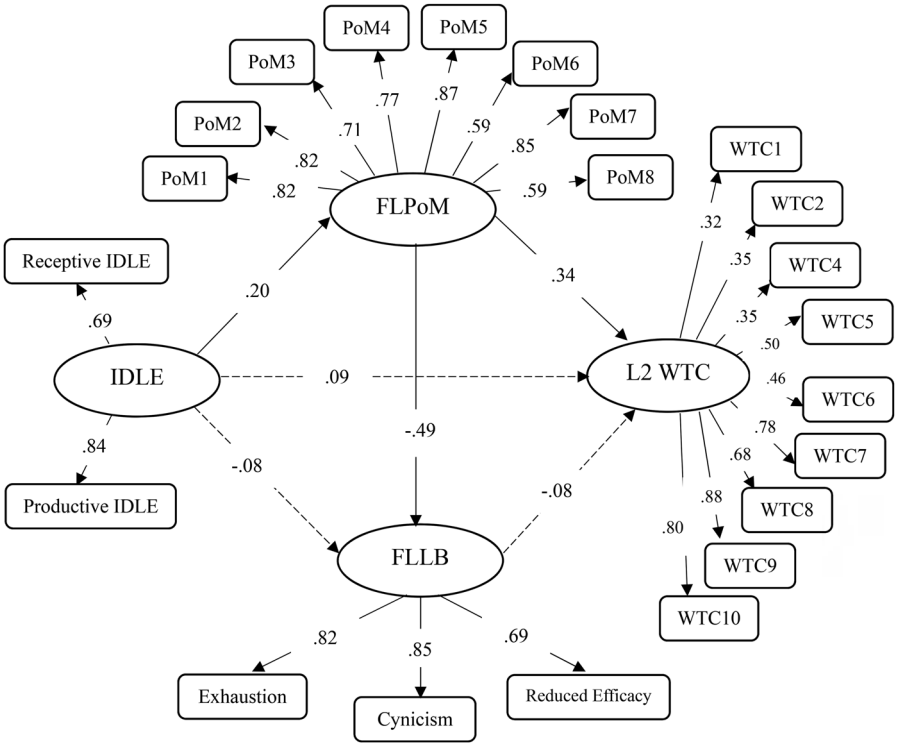


Figure 2. The mediation model.
Note. The dashed lines show non-significant paths ($p > .01$).

Table 5. Mediation model results.

	Estimate	*p*-value
Direct effects		
IDLE → L2WTC	.165	.025
IDLE → FLPoM	.196	.010
IDLE → FLLB	−0.176	.046
FLPoM → L2 WTC	.375	.001
FLLB → L2 WTC	−0.078	.327
FLPoM → FLLB	−0.486	.001
Indirect effects		
IDLE → FLPoM → L2 WTC	.032	.008
IDLE → FLPoM → FLLB	−0.077	.009
IDLE → FLPoM → FLLB → L2 WTC	.004	.145
IDLE → FLLB → L2 WTC	.003	.220

= .025) and indirectly through FLLB (β = .003, p = .220). The 95% confidence interval of the indirect effect of IDLE on L2 WTC through FLLB did not contain zero (CI [–0.002; .027]), further confirming the absence of a mediation effect.

In addition, large direct effects were found between IDLE and FLPoM (β = .196; p = .010) and between FLPoM and FLLB (β = −0.486; $p <$ .01). As the direct effect of IDLE on L2 WTC was not statistically significant, only FLPoM fully mediated the relationship between IDLE and L2 WTC and between IDLE and FLLB.

Discussion

The present study was designed to investigate the mediating role of FLPoM and FLLB in the relationship between EFL learners' IDLE and L2 WTC. The proposed mediation model depicted that IDLE did not directly impact language learners' WTC. In addition, based on the results, FLPoM not only played a mediating role between EFL learners' IDLE and L2 WTC but also mediated the relationship between IDLE and FLLB. From a theoretical viewpoint, the significant direct influence of FLPoM on language learners' FLLB and WTC is in line with the tenets of positive psychology, which highlights the crucial role of positive psychological states in language education (Derakhshan et al., 2022; Dewaele et al., 2023). In addition, the findings of the current study are in support of the broaden-and-build theory which postulates the effects of positive variables on broadening a person's momentary thought-action tendencies and expanding their enduring psychological and cognitive resources (Fredrickson, 2013). In support of Badiei et al. (2023) and Zhou et al. (2023), the current study highlighted the positive role of FLPoM in affecting students' linguistics outcomes. The findings also corroborate with the idea of Ebn-Abbasi et al. (2024) suggesting that positive individual features can promote EFL learners' WTC.

An interesting finding of this research study was that EFL learners' IDLE was not directly associated with their L2 WTC and FLLB. On the one hand, it was revealed that FLPoM had a full mediation role in the relationship between EFL learners' IDLE and L2 WTC; this means IDLE could not directly lead to L2 WTC. This finding contrasts with some studies that have reported that the facilitative nature of digital settings can enhance L2 WTC (Peterson, 2016; Reinders & Wattana, 2015). Furthermore, it is in contrast with several studies which found that EFL learners who engage in a range of IDLE activities are more willing to communicate in L2 in the contexts of Indonesia (Lee & Drajati, 2019; Nugroho, 2021), Korea (Lee & Dressman, 2018; Lee & Sylvén, 2021), and Taiwan (Lee et al., 2019). This inconsistency, as Khajavy et al. (2016) stated, may be due to various educational factors (e.g. classroom environment, student cohesiveness, and teacher support) influencing EFL learners' WTC. In particular, it has been shown that different interpersonal variables (e.g. supportive online communities) and affective factors (e.g. L2 confidence) influence Iranian EFL learners' WTC (Soyoof, 2023). Nevertheless, this finding is in line with the findings of some other studies which have shown, for example, that mobile language learning in informal environments produces limited effects (Burston, 2014; Sung et al., 2015). This is also supported by Lee and Sylvén (2021), who revealed that productive IDLE and receptive

IDLE in Sweden were not significant predictors of L2 WTC. More importantly, this result is consistent with Zadorozhnyy and Lee (2023) who found receptive and productive IDLE were not directly associated with L2 WTC in the context of Kazakhstan and EFL Kazakhstani learners who engaged in IDLE did not necessarily become willing to communicate in English. The current literature on IDLE has provided mixed results (Lee, 2019) proving that research on IDLE is still in its infancy (Lee, 2019; Lee & Lee, 2021; Soyoof et al., 2023) and there is a need for more extensive research in a variety of contexts. Additionally, some typical IT-related problems exist in the context of Iran which could have contributed to this result. One of them refers to the poor internet connection in some regions of the nation, causing IDLE practices to be inconvenient for some Iranian EFL learners. Furthermore, some online platforms (e.g. Instagram or YouTube) are restricted nationwide, which decreases EFL learners' opportunities to access IDLE environments easily. Consequently, such limitations and discomfort might have resulted in no direct impact of IDLE on Iranian EFL learners' L2 WTC.

On the other hand, this study also found that IDLE had an indirect influence on learners' FLLB through the full mediating role of FLPoM. This implied that IDLE could not directly prevent Iranian EFL learners from experiencing FLLB. This particular finding is in contrast to Lee and Drajati (2019) who found IDLE positively correlates with a lack of negative feelings and anxiety. However, this result gives credence to Li et al. (2021) who found FLLB is a context-specific state and even fully engaged learners in a subject may occasionally experience it, especially before exams. Most of our participants, who were university or school students, participated in the study in June (before their final exams). This might have resulted in them reporting high levels of burnout.

Another finding of this study was a significant relationship between Iranian EFL learners' IDLE and their FLPoM, suggesting that digital learning of English outside the classroom triggers EFL learners' positive feelings. This finding corroborates the results of previous studies (e.g. Lai et al., 2015) suggesting a positive relationship between IDLE-based activities and EFL learners' positive emotions. Additionally, this outcome is closely aligned with the study of Lee et al. (2024) showing that IDLE practices predict a higher level of positive L2 emotion (i.e. enjoyment) and a lower level of anxiety. Moreover, this finding is in line with Nugroho and Atmojo (2020) who found that EFL learners positively perceived the use of digital technology as a means of language learning, and Lee and Lee (2021) who concluded that IDLE practices predicted EFL learners' L2 enjoyment.

This study also showed that FLPoM positively mediated the relationship between IDLE and L2 WTC. This finding aligns with the results of

previous studies that showed a state of peace and harmony positively impacts learners' academic outcomes, including academic motivation, and engagement (Datu, 2017; Datu et al., 2018). Furthermore, this finding is consistent with several studies that have revealed that positive emotions could play a positive role in EFL learners' L2 WTC (Khajavy et al., 2018) and IDLE (Lee & Lee, 2021). It is also consistent with Lee (2019) who reported that a lack of anxiety in EFL learning was positively associated with L2 WTC and IDLE. Moreover, this finding is in line with Lee et al. (2024) and Lee et al. (2022) who identified that positive emotions (i.e. L2 enjoyment) partially mediated the relationship between IDLE and L2 WTC.

Last but not least, the results revealed that FLPoM mediated the relationship between IDLE and FLLB. This finding shows that FLPoM can lead language learners to experience low levels of learning burnout. This aligns with the results of MacIntyre et al.'s (2019) empirical investigation, according to which the development of positive emotions can undo the effects of negative feelings and equip EFL learners with resources to recover from them. It is also consistent with Zhou et al. (2023) who uncovered that the effects of negative learner variables could be countered by making EFL learners experience positive feelings. As stated by Zhou et al. (2023), FLPoM helps EFL learners 'build and conserve individuals' psychological resources, such as psychological resilience or self-efficacy, because a stable and balanced state of mind encourages individuals to play down external stimuli, especially negative ones and protects them against stress or psychological strain' (p. 8).

In sum, this study showed that FLPoM mediated not only the relationship between IDLE and L2 WTC, but also the relationship between IDLE and FLLB in the context of Iran. In other words, IDLE activities will enhance learners' L2 WTC if they are accompanied by experiencing FLPoM. Also, if FLPoM accompanies the learners' IDLE practices, their FLLB will decrease. We argue that understanding EFL learners' positive psychological states and negative psychological syndromes is fundamental since they directly affect their L2 performance (Zhou et al., 2023) both inside and outside the classroom settings. Therefore, EFL learning can flourish when learners value and experience FLPoM in their language learning processes.

Limitations and suggestions for future research

The current research study is not without limitations. First, this study only utilized self-report data; further qualitative methods are required to investigate the role of positive and negative psychosocial states in informal, digital English learning so that a more comprehensive picture of

this intersection would be gained. Future studies can employ unstructured interviews to allow for more open-ended discussions of participants' voices and perspectives. Moreover, journaling can be used as another data collection tool to generate detailed data capturing nuances, complexities, and subjective experiences that may not be captured through quantitative approaches. Second, gender fell outside the scope of our research as the majority of our participants were females; further research is needed to explore the associations among the variables with a homogeneous sample in terms of gender since it may play a role. Third, as both FLPoM and FLLB are novel and under-explored variables, more studies with a similar focus are required to be conducted in different EFL contexts to verify the proposed model. Finally, researchers can expand current efforts by means of studying informal learning communities that can mediate the relationship between IDLE and L2 WTC such as Discord, Linguaholic, DuoLingo, Rosetta Stone, and Second Life (e.g. Odinokaya et al., 2021; Shortt et al., 2023).

Conclusion and implications

The present study showed that FLPoM mediated the relationship between Iranian EFL learners' IDLE and L2 WTC. Also, the results showed that FLPoM was a total mediator of the relationship between EFL learners' IDLE and FLLB. The results highlighted the significant role of positive emotions in the process of digital EFL learning. By demonstrating that EFL learners' FLPoM can lead to an increase in their L2 WTC and a decrease in their FLLB, we propose a more comprehensive perception of how positive emotional states influence and facilitate language learning processes in out-of-class settings through digital technologies. An implication is that EFL learners' out-of-class, digital learning can also hinge upon a state of peace and harmony they experience.

EFL teachers are recommended to increase peace of mind and harmony among their learners, whether inside or outside of class, to facilitate their L2 achievements. To help learners achieve a state of harmony inside the classroom, teachers can foster collaboration among learners by creating a positive learning environment, establishing a classroom culture valuing collaboration, and encouraging learners' use of peer feedback. In addition to introducing user-friendly mobile applications and practical websites, EFL teachers should boost their learners' confidence in practicing language abilities through online platforms with no fear of being judged. One way to help language learners achieve a sense of peace of mind in digital language learning in out-of-class settings is creating communities of inquiry and synchronous chat forums which not only can enhance language skills, but also can create a supportive learning

environment. These communities can be available in online forums or social media groups. The benefits of such communities include sharing useful resources (e.g. language learning websites, videos, or podcasts) that can facilitate learners' WTC in L2, encouraging peer-to-peer feedback to overcome learners' language inefficiency, creating a welcoming environment where every single member feels supported in their learning process, and celebrating individual achievements within the community to prevent a sense of burnout.

Creating a supportive atmosphere in online language learning settings can be another helpful strategy for enhancing the possibility of communicating in English (Börekci & Aydin, 2020). This can be achieved by fostering a sense of community among learners and providing opportunities to connect and interact in L2 through digital and online platforms. According to Vygotsky and Cole's (1978) sociocultural theory, learning hinges to a large extent on the social collaboration between the learners and their peers (Lantolf & Beckett, 2009). Such collaborative chances can be constituted within productive online language learning communities (Simpson, 2005), where computers and digital platforms can act as mediational tools in collaborative activity (Gánem Gutiérrez, 2006). In addition, using digital platforms that arouse a sense of enjoyment and a state of harmony among foreign language learners is highly recommended as they help learners communicate in L2. Digital material developers can consider this; learners react emotionally to the digital, informal English materials. Accordingly, after a needs analysis, a wide range of materials should be prepared based on the learners' individual differences and personal preferences. Furthermore, teachers can ask the students to share their own experiences of practicing L2 in informal digital learning environments and the available software or platforms they utilize in this regard, so that their classmates would benefit from such peer experiences. Also, teachers can constantly upgrade their own knowledge base on new IDLE platforms, and recommend useful ones in terms of L2 learning to the students. Moreover, teachers can take the responsibility of teaching students how to regulate and navigate their emotions in IDLE contexts as well as how to respond to failures in such environments. From a pedagogical perspective, EFL teachers are recommended to help their students create a personalized IDLE environment to be able to practice English in out of formal settings (Toffoli et al., 2023). EFL teachers can capitalize on the experiences of the students who have successfully use IDLE activities (Lee, 2021). Therefore, having small talks with these learners, EFL teachers can learn about various digital resources and materials, technological tools, language learning software, and successful online language learning strategies. In addition, EFL teachers can have one-to-one or small group meetings with their students to provide them with individualized

metacognitive strategies by explaining how digital resources should be used independent of classroom learning. The final message is clear: Positive and negative emotional and psychological states are everywhere; they are not only contagious inside foreign language classes but also are determinant factors in the digital world, outside the classroom setting.

Acknowledgements

The authors sincerely thank all the EFL learners who participated in the study.

Ethical approval

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Authors' contribution

All authors have materially participated in the research and article preparation. In addition, all authors have approved the final article.

Consent to participate

Informed consent to participate was obtained from all individual participants included in the study.

Consent for publication

Informed consent for publication was obtained from all individual participants included in the study.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

References

Alrabai, F. (2022). Modeling the relationship between classroom emotions, motivation, and learner willingness to communicate in EFL: Applying a holistic approach of positive psychology in SLA research. *Journal of Multilingual and Multicultural Development*, Advance online publication. https://doi.org/10.1080/01434632.2022.2053138

Azari Noughabi, M., & Ghasemi, A. (2024). Informal digital learning of English and EFL learners' willingness to communicate: Investigating the mediating role of L2 grit. *Journal of Multilingual and Multicultural Development*, Advance online publication. https://doi.org/10.1080/01434632.2024.2351087

Badiei, G., Azari Noughabi, M., & Amirian, S. K. (2023). Foreign language peace of mind, foreign language enjoyment, and perceived language proficiency: The mediating role of L2 grit. *English Teaching and Learning*. Advance online publication. https://doi.org/10.1007/s42321-023-00163-8

Benson, P. (2011). Language learning and teaching beyond the classroom: An introduction to the field. In P. Benson & H. Reinders (Eds.), *Beyond the language classroom* (pp. 7–16). Palgrave Macmillan.

Biesanz, J. C., Falk, C. F., & Savalei, V. (2010). Assessing mediational models: Testing and interval estimation for indirect effects. *Multivariate Behavioral Research*, *45*(4), 661–701. https://doi.org/10.1080/002

Börekci, R., & Aydin, S. (2020). Foreign language teachers' interactions with their students on Facebook. *Computer Assisted Language Learning*, *33*(3), 217–239. https://doi.org/10.1080/09588221.2018.1557691

Burston, J. (2014). MALL: The pedagogical challenges. *Computer Assisted Language Learning*, *27*(4), 344–357. https://doi.org/10.1080/09588221.2014.914539

Cameron, D. (2021). Case studies of Iranian migrants' WTC within an ecosystems framework: The influence of past and present language learning experiences. In N. Zarrinabadi, & M. Pawlak (Eds.), *New perspectives on willingness to communicate in a second language* (pp. 25–53). Springer International Publishing. https://doi.org/10.1007/978-3-030-67634-6_3

Chen, W., Sun, P., Zhang, A., & Yang, Z. (2023). How does foreign language peace of mind contribute to learning performance: Roles of academic engagement and social

support. *Journal of Multilingual and Multicultural Development*, Advance online publication. https://doi.org/10.1080/01434632.2023.2283194

Chu, H. R. (2008). *Shyness and EFL learning in Taiwan: A study of shy and non-shy college students' use of strategies, foreign language anxiety, motivation, and willingness to communicate* [Unpublished PhD dissertation]. University of Texas.

Datu, J. A. D. (2017). Peace of mind, academic motivation, and academic achievement in Filipino high school students. *The Spanish Journal of Psychology*, *20*, E22. https://doi.org/10.1017/sjp.2017.19

Datu, J. A. D., Valdez, J. P. M., & King, R. B. (2018). Exploring the association between peace of mind and academic engagement: Cross-sectional and cross-lagged panel studies in the Philippine context. *Journal of Happiness Studies*, *19*, 1903–1916. https://doi.org/10.1007/s10902-017-9902-x

Derakhshan, A., Dewaele, J. M., & Azari Noughabi, M. (2022). Modeling the contribution of resilience, well-being, and L2 grit to foreign language teaching enjoyment among Iranian English language teachers. *System*, *109*, 102890. https://doi.org/10.1016/j.system.2022.102890

Dewaele, J., Botes, E., & Greiff, S. (2023). Sources and effects of foreign language enjoyment, anxiety, and boredom: A structural equation modeling approach. *Studies in Second Language Acquisition*, *45*(2), 461–479. https://doi.org/10.1017/S0272263122000328

Dewaele, J. M., & MacIntyre, P. D. (2014). The two faces of Janus? Anxiety and enjoyment in the foreign language classroom. *Studies in Second Language Learning and Teaching*, *4*(2), 237–274. https://doi.org/10.14746/ssllt.2014.4.2.5

Dewaele, J. M., & Meftah, R. (2023). The trajectory of English foreign learners' emotions and motivation from the start to the end of their learning journey: A pseudo-longitudinal investigation. *Journal of the European Second Language Association*, *7*(1), 16–30. https://doi.org/10.22599/jesla.94

Dewaele, J.-M., Saito, K., & Halimi, F. (2022). How teacher behaviour shapes foreign language learners' enjoyment, anxiety and motivation: A mixed modelling longitudinal investigation. *Language Teaching Research*, Advance online publication. https://doi.org/10.1177/13621688221089601

Dressman, M. (2019). Informal English learning among Moroccan youth. In M. Dressman & R. W. Sadler (Eds.), *The handbook of informal language learning* (pp. 303–318). John Wiley & Sons Ltd. https://doi.org/10.1002/9781119472384.ch20

Ebn-Abbasi, F., Fattahi, N., Azari Noughabi, M., & Botes, E. (2024). The strength of self and L2 willingness to communicate: The role of L2 grit, ideal L2 self and language mindset. *System*, *123*, 103334. https://doi.org/10.1016/j.system.2024.103334

Ellis, N. C., & Wulff, S. (2014). Usage-based approaches to SLA. In B. Van Patten & J. Williams (Eds.), *Theories in second language acquisition: An introduction* (pp. 87–105). Routledge.

Feng, E., Wang, Y., & King, R. B. (2023). Achievement goals, emotions and willingness to communicate in EFL learning: Combining variable-and person-centered approaches. *Language Teaching Research*, Advance online publication. https://doi.org/10.1177/13621688221146887

Field, A. (2013). *Discovering statistics using IBM SPSS statistics*. SAGE.

Fredrickson, B. L. (2001). The role of positive emotions in positive psychology: The broaden and build theory of positive emotions. *American Psychologist*, *56*(3), 218–226. https://doi.org/10.1037/0003-066x.56.3.218

Fredrickson, B. L. (2013). Positive emotions broaden and build. In P. Devine & A. Plant (Eds.), *Advances in experimental social psychology* (pp. 1–54). Academic Press.

Fuente, M. J. (2003). Is SLA interactionist theory relevant to CALL? A study on the effects of computer-mediated interaction in L2 vocabulary acquisition. *Computer Assisted Language Learning*, *16*(1), 47–81. https://doi.org/10.1076/call.16.1.47.15526

Gánem Gutiérrez, G. (2006). Sociocultural theory and its application to CALL: A study of the computer and its relevance as a mediational tool in the process of collaborative activity. *ReCALL*, *18*(2), 230–251. https://doi.org/10.1017/S0958344006000620

Gass, S. M., & Mackey, A. (2014). Input, interaction, and output in second language acquisition. In B. Van Patten & J. Williams (Eds.), *Theories in second language acquisition: An introduction* (pp. 194–220). Routledge.

Gass, S. M., & Mackey, A. (2020). Input, interaction, and output in L2 acquisition. In B. Van Patten, G. D. Keating, & S. Wulff (Eds.), *Theories in second language acquisition: An introduction* (pp. 192–222). Routledge.

Gass, S. M., Behney, J., & Plonsky, L. (2020). *Second language acquisition: An introductory course*. Routledge.

Ghonsooly, B., Khajavy, G. H., & Asadpour, S. F. (2012). Willingness to communicate in English among Iranian non–English major university students. *Journal of Language and Social Psychology*, *31*(2), 197–211. https://doi.org/10.1177/0261927X12438538

Hu, L. T., & Bentler, P. (1995). Evaluating model fit. In R. H. Hoyle (Ed.), *Structural equation modeling: Concepts, issues, and applications* (pp. 76–99). Sage Publications.

Khajavy, G. H., Ghonsooly, B., Hosseini Fatemi, A., & Choi, C. W. (2016). Willingness to communicate in English: A microsystem model in the Iranian EFL classroom context. *TESOL Quarterly*, *50*(1), 154–180. https://doi.org/10.1002/tesq.204

Khajavy, G. H., MacIntyre, P. D., & Barabadi, E. (2018). Role of the emotions and classroom environment in willingness to communicate: Applying doubly latent multilevel analysis in second language acquisition research. *Studies in Second Language Acquisition*, *40*(3), 605–624. https://doi.org/10.1017/S0272263117000304

Khajavy, G. H., MacIntyre, P. D., Taherian, T., & Ross, J. (2021). Examining the dynamic relationships between willingness to communicate, anxiety and enjoyment using the experience sampling method. In N. Zarrinabadi & M. Pawlak (Eds.). *New perspectives on willingness to communicate in a second language* (pp. 169–197). Springer International Publishing.

Kline, R. B. (2015). The mediation myth. *Basic and Applied Social Psychology*, *37*(4), 202–213. https://doi.org/10.1080/01973533.2015.1049349

Kruk, M., Pawlak, M., Shirvan, M. E., Taherian, T., & Yazdanmehr, E. (2022). Potential sources of foreign language learning boredom: AQ methodology study. *Studies in Second Language Learning and Teaching*, *12*(1), 37–58. https://doi.org/10.14746/ssllt.2022.12.1.3

Kumpikaitė-Valiūnienė, V., Aslan, I., Duobienė, J., Glińska, E., & Anandkumar, V. (2021). Influence of digital competence on perceived stress, burnout and well-being among students studying online during the COVID-19 lockdown: A 4-country perspective. *Psychology Research and Behavior Management*, *14*, 1483–1498. https://doi.org/10.2147/PRBM.S325092

Kun, Y., Senom, F., & Peng, C. F. (2020). Relationship between willingness to communicate in English and foreign language enjoyment. *Universal Journal of Educational Research*, *8*(10), 4853–4862. https://doi.org/10.13189/ujer.2020.081057

Lai, C., Zhu, W., & Gong, G. (2015). Understanding the quality of out-of-class English learning. *TESOL Quarterly*, *49*(2), 278–308. https://doi.org/10.1002/tesq.171

Lantolf, J. P., & Beckett, T. G. (2009). Sociocultural theory and second language acquisition. *Language Teaching*, *42*(4), 459–475. https://doi.org/10.1017/S0261444809990048

Lee, J. S. (2019). Informal digital learning of English and second language vocabulary outcomes: Can quantity conquer quality? *British Journal of Educational Technology*, *50*(2), 767–778. https://doi.org/10.1111/bjet.12599

Lee, J. S. (2021). *Informal digital learning of English: Research to practice*. Routledge.

Lee, J. S., & Chiu, M. M. (2023). Modeling EFL learners' willingness to communicate: The roles of face-to-face and digital L2 communication anxiety. *Annual Review of Applied Linguistics*, *43*, 64–87. https://doi.org/10.1017/S0267190523000090

Lee, J. S., & Drajati, N. A. (2019). English as an international language beyond the ELT classroom. *ELT Journal*, *73*(4), 419–427. https://doi.org/10.1093/elt/ccz018

Lee, J. S., & Dressman, M. (2018). When IDLE hands make an English workshop: Informal digital learning of English and language proficiency. *TESOL Quarterly*, *52*(2), 435–445. https://doi.org/10.1002/tesq.422

Lee, J. S., & Hsieh, J. C. (2019). Affective variables and willingness to communicate of EFL learners in in-class, out-of-class, and digital contexts. *System*, *82*, 63–73. https://doi.org/10.1016/j.system.2019.03.002

Lee, J. S., & Lee, K. (2021). The role of informal digital learning of English and L2 motivational self-system in foreign language enjoyment. *British Journal of Educational Technology*, *52*(1), 358–373. https://doi.org/10.1111/bjet.12955

Lee, J. S., Lee, K., & Hsieh, J. C. (2019). Understanding willingness to communicate in L2 between Korean and Taiwanese students. *Language Teaching Research*, *26*(3), 455–476. https://doi.org/10.1177/1362168819890825Scale

Lee, J. S., & Sylvén, L. K. (2021). The role of informal digital learning of English in Korean and Swedish EFL learners' communication behavior. *British Journal of Educational Technology*, *52*(3), 1279–1296. https://doi.org/10.1111/bjet.13082

Lee, J. S., Xie, Q., & Lee, K. (2024). Informal digital learning of English and L2 willingness to communicate: Roles of emotions, gender, and educational stage. *Journal of Multilingual and Multicultural Development*, *45*(2), 596–612. https://doi.org/10.1080/01434632.2021.1918699

Lee, J. S., Yeung, N. M., & Osburn, M. B. (2022). Foreign language enjoyment as a mediator between informal digital learning of English and willingness to communicate: A sample of Hong Kong EFL secondary students. *Journal of Multilingual and Multicultural Development*, Advance online publication. https://doi.org/10.1080/01434632.2022.2112587

Lee, Y. C., Lin, Y. C., Huang, C. L., & Fredrickson, B. L. (2013). The construct and measurement of peace of mind. *Journal of Happiness Studies*, *14*, 571–590. https://doi.org/10.1007/s10902-012-9343-5

Li, C. (2020). Emotional intelligence and English achievement: The mediating effects of enjoyment, anxiety and burnout. *Foreign Language World*, *1*, 69–78.

Li, C., Zhang, L. J., & Jiang, G. (2021). Conceptualisation and measurement of foreign language learning burnout among Chinese EFL students. *Journal of Multilingual and Multicultural Development*, Advance online publication. https://doi.org/10.1080/01434632.2021.1931246

Liu, C., He, J., Ding, C., Fan, X., Hwang, G. J., & Zhang, Y. (2021). Self-oriented learning perfectionism and English learning burnout among EFL learners using mobile applications: The mediating roles of English learning anxiety and grit. *Learning and Individual Differences*, *88*, 102011. https://doi.org/10.1016/j.lindif.2021.102011

Liu, G., Zhang, Y., & Zhang, R. (2023). Bridging imagination and informal digital learning of English: A mixed-method investigation. *Journal of Multilingual and Multicultural Development*, Advance online publication. https://doi.org/10.1080/01434632.2023.2173214

Long, M. H. (1996). The role of linguistic environment in second language acquisition. In W. C. Ritchie & T. K. Bhatia (Eds.), *Handbook of second language acquisition* (pp. 413–468). Academic Press.

MacIntyre, P. D., Clément, R., Dörnyei, Z., & Noels, K. A. (1998). Conceptualizing willingness to communicate in a L2: A situational model of L2 confidence and affiliation. *Modern Language Journal*, *82*(4), 545–562. https://doi.org/10.1111/j.1540-4781.1998.tb05543.x

MacIntyre, P. D., Gregersen, T., & Mercer, S. (2019). Setting an agenda for positive psychology in SLA: Theory, practice, and research. *The Modern Language Journal*, *103*(1), 262–274. https://doi.org/10.1111/modl.12544

MacIntyre, P. D., & Vincze, L. (2017). Positive and negative emotions underlie motivation for L2 learning. *Studies in Second Language Learning and Teaching*, *7*(1), 61–88. https://doi.org/10.14746/ssllt.2017.7.1.4

MacIntyre, P. D., & Wang, L. (2021). Willingness to communicate in the L2 about meaningful photos: Application of the pyramid model of WTC. *Language Teaching Research*, *25*(6), 878–898. https://doi.org/10.1177/13621688211004645

Maslach, C., Schaufeli, W. B., & Leiter, M. P. (2001). Job burnout. *Annual Review of Psychology*, *52*(1), 397–422. https://doi.org/10.1146/annurev.psych.52.1.397

McCroskey, J. C., & Baer, J. E. (1985). *Willingness to communicate: The construct and its measurement* [Paper presentation]. The annual meeting of the Speech Communication Association. the United States.

Mitchell, R., Myles, F., & Marsden, E. (2013). *Second language learning theories* (3rd ed.). Routledge.

Niu, L., Wang, X., Wallace, M. P., Pang, H., & Xu, Y. (2022). Digital learning of English as a foreign language among university students: How are approaches to learning linked to digital competence and technostress? *Journal of Computer Assisted Learning*, *38*(5), 1332–1346. https://doi.org/10.1111/jcal.12679

Nugroho, A. (2021). Learners willingness to communicate in a foreign language: The role of informal digital learning of English. *English Language Education Reviews*, *1*(2), 125–133. https://doi.org/10.22515/ele-reviews.v1i2.4396

Nugroho, A., & Atmojo, A. E. P. (2020). Digital learning of English beyond classroom: EFL learners' perception and teaching activities. *Journal of English Education and Linguistics Studies*, *7*(2), 219–243. https://doi.org/10.30762/jeels.v7i2.1993

Odinokaya, M. A., Krylova, E. A., Rubtsova, A. V., & Almazova, N. I. (2021). Using the Discord application to facilitate EFL vocabulary acquisition. *Education Sciences*, *11*, 470. https://doi.org/10.3390/educsci11090470

Pawlak, M., Kruk, M., & Zawodniak, J. (2022). Investigating individual trajectories in experiencing boredom in the language classroom: The case of 11 Polish students of English. *Language Teaching Research*, *26*(4), 598–616. https://doi.org/10.1177/1362168820914004

Pawlak, M., Zawodniak, J., & Kruk, M. (2020). *Boredom in the foreign language classroom: A micro-perspective*. Springer Nature.

Peng, J., & Woodrow, L. (2010). Willingness to communicate in English: A model in the. Chinese EFL classroom context. *Language Learning*, *60*(4), 834–876. https://doi.org/10.1111/j.1467-9922.2010.00576.x

Peterson, M. (2016). The use of massively multiplayer online role-playing games in CALL: An analysis of research. *Computer Assisted Language Learning*, *29*(7), 1181–1194. https://doi.org/10.1080/09588221.2016.1197949

Plonsky, L., & Oswald, F. L. (2014). How big is 'Big'? Interpreting effect sizes in L2 research. *Language Learning*, *64*(4), 878–912. https://doi.org/10.1111/lang.12079

Reinders, H., & Wattana, S. (2015). Affect and willingness to communicate in digital game-based learning. *ReCALL*, *27*(1), 38–57. https://doi.org/10.1017/S0958344014000226

Richards, J. C. (2015). The changing face of language learning: Learning beyond the classroom. *RELC Journal*, *46*(1), 5–22. https://doi.org/10.1177/0033688214561621

Shao, K., Pekrun, R., & Nicholson, L. J. (2019). Emotions in classroom language learning: What can we learn from achievement emotion research? *System*, *86*, 102121. https://doi.org/10.1016/j.system.2019.102121

Shortt, M., Tilak, S., Kuznetcova, I., Martens, B., & Akinkuolie, B. (2023). Gamification in mobile-assisted language learning: A systematic review of Duolingo literature from public release of 2012 to early 2020. *Computer Assisted Language Learning*, *36*(3), 517–554. https://doi.org/10.1080/09588221.2021.1933540

Simpson, J. (2005). Learning electronic literacy skills in an online language learning community. *Computer Assisted Language Learning*, *18*(4), 327–345. https://doi.org/10.1080/09588220500335463

Soyoof, A. (2023). Iranian EFL students' perception of willingness to communicate in an extramural digital context. *Interactive Learning Environments*, *31*(9), 5922–5939. https://doi.org/10.1080/10494820.2021.2024579

Soyoof, A., Reynolds, B. L., Vazquez-Calvo, B., & McLay, K. (2023). Informal digital learning of English (IDLE): A scoping review of what has been done and a look towards what is to come. *Computer Assisted Language Learning*, *36*(4), 608–640. https://doi.org/10.1080/09588221.2021.1936562

Sun, Y., Franklin, T., & Gao, F. (2017). Learning outside of classroom: Exploring the active part of an informal online English learning community in China. *British Journal of Educational Technology*, *48*, 57–70. https://doi.org/10.1111/bjet.12340

Sung, Y. T., Chang, K. E., & Yang, J. M. (2015). How effective are mobile devices for language learning? A meta-analysis. *Educational Research Review*, *16*, 68–84. https://doi.org/10.1016/j.edurev.2015.09.001

Swain, M. (1985). Communicative competence: Some roles of comprehensible input and comprehensible output in its development. *Input in Second Language Acquisition*, *15*, 165–179.

Swain, M. (1995). Three functions of output in second language learning. In G. Cook, & B. Seidlhofer (Eds.), *Principle and practice in applied linguistics: Studies in Honour of H. G. Widdowson* (pp. 125–144). Oxford University Press.

Taherian, T., Shirvan, M. E., Yazdanmehr, E., Kruk, M., & Pawlak, M. (2023). A longitudinal analysis of informal digital learning of English, willingness to communicate and foreign language Boredom: A latent change score mediation model. *The Asia-Pacific Educatio Researcher*, Advance online publication. https://doi.org/10.1007/s40299-023-00751-z

Teimouri, Y. (2016). L2 selves, emotions, and motivated behaviors. *Studies in Second Language Acquisition*, *39*(4), 681–709. https://doi.org/10.1017/S0272263116000243

Toffoli, D., Sockett, G., & Kusyk, M. (Eds.). (2023). *Language learning and leisure: Informal language learning in the digital age*. De Gruyter Mouton. https://doi.org/10.1515/9783110752441

Van Patten, B., Keating, G.D., & Wulff, S. (Eds.). (2020). *Theories in second language acquisition: An introduction* (3rd ed.) Routledge. https://doi.org/10.4324/9780429503986

Vizoso, C., Arias-Gundín, O., & Rodríguez, C. (2019). Exploring coping and optimism as predictors of academic burnout and performance among university students. *Educational Psychology*, *39*(6), 768–783. https://doi.org/10.1080/01443410.2018.1545996

Vygotsky, L. S., & Cole, M. (1978). *Mind in society: Development of higher psychological processes*. Harvard University Press.

Wongsa, M., & Son, J. B. (2022). Enhancing Thai secondary school students' English speaking skills, attitudes and motivation with drama-based activities and Facebook. *Innovation in Language Learning and Teaching*, *16*(1), 41–52. https://doi.org/10.1080/17501229.2020.1853134

Yang, S., Azari Noughabi, M., Botes, E., & Dewaele, J. M. (2023). Let's get positive: How foreign language teaching enjoyment can create a positive feedback loop. *Studies in Second Language Learning and Teaching*, *13*(1), 17–38. https://doi.org/10.14746/ssllt.32358

Yashima, T. (2019). L2 motivation and willingness to communicate. In M. Lamb, K. Csizér, A. Henry, & S. Ryan (Eds.), *The Palgrave handbook of motivation for language learning* (pp. 203–223). https://doi.org/10.1007/978-3-030-28380-3_10

Yen, Y. C., Hou, H. T., & Chang, K. E. (2015). Applying role-playing strategy to enhance learners' writing and speaking skills in EFL courses using Facebook and Skype as learning tools: A case study in Taiwan. *Computer Assisted Language Learning*, *28*(5), 383–406. https://doi.org/10.1080/09588221.2013.839568

Wang, H., Peng, A., & Patterson, M. M. (2021). The roles of class social climate, language mindset, and emotions in predicting willingness to communicate in a foreign language. *System*, *99*, 102529. https://doi.org/10.1016/j.system.2021.102529

Zadorozhnyy, A., & Lee, J. S. (2023). Informal digital learning of English and willingness to communicate in a second language: Self-efficacy beliefs as a mediator. *Computer Assisted Language Learning*, Advance online publication. https://doi.org/10.1080/09588221.2023.2215279

Zarrinabadi, N., Khodarahmi, E., & Shahbazi, H. (2021). On the effect of using a flipped classroom methodology on Iranian EFL learners' willingness to communicate. In N. Zarrinabadi, & M. Pawlak (Eds.), *New Perspectives on willingness to communicate in a second language* (pp. 155–167). Springer International Publishing. https://doi.org/10.1007/978-3-030-67634-6_8

Zhang, Y., & Liu, G. (2022). Revisiting informal digital learning of English (IDLE): A structural equation modeling approach in a university EFL context. *Computer Assisted Language Learning*, Advance online publication. https://doi.org/10.1080/09588221.2022.2134424

Zhou, L., Dewaele, J. M., Lochtman, K., & Xi, Y. (2023). Foreign language peace of mind: A positive emotion drawn from the Chinese EFL learning context. *Applied Linguistics Review*, *14*(5), 1385–1410. https://doi.org/10.1515/applirev-2021-0080